Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Computer Science/ Computer Systems Engineering

Final Examination
Year 2, Semester 1 (2025)

SE2032 – Database Management Systems

Duration: 3 Hours

November 2025

Instructions to Candidates:

- This paper has 5 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 8 pages, including the cover page.
- Scientific calculators are allowed. Any other devices capable of storing and retrieving text, including mobile phones are not allowed.

Question 1 (20 Marks)

A database designer is designing a database for a Hospital Patient Management System. The following requirements have been gathered:

At a hospital, doctors lead medical treatments and oversee patient care. Each doctor has a unique DoctorID, Name, Email, and Specialization. Doctors supervise multiple treatments. A treatment has a unique TreatmentID, Description, StartDate, and Cost. Each treatment must be supervised by exactly one doctor, but a doctor may supervise multiple treatments.

Every patient has a PatientID, Name, Email, and InsuranceType. A patient can undergo multiple treatments, and in each, they are assigned a specific role (e.g., Primary Patient, Secondary Patient). Additionally, patients may act as health mentors for other patients in wellness programs. Patients can be further categorized into Inpatients (admitted to the hospital) and Outpatients (visiting for consultation without admission) with attributes specific to their category: Inpatients may have RoomNumber and AdmissionDate, while Outpatients may have NextAppointmentDate. The prescription includes attributes such as PrescriptionNumber, Dosage, Frequency, and Duration. The hospital records which doctor issued each prescription for a particular patient with ApprovalDate indicating when the doctor authorized the prescription.

For every treatment, a number of TestReports are prepared to record the diagnostic results. Each TestReport depends on a single Treatment. It is identified by a ReportNumber that is unique only within the context of its Treatment. The TestReport include ReportNumber, TestType, Result, and TestDate.

Construct an EER Diagram that includes:

(a) All relevant entities and their attributes. **(6 marks)**

(b) Appropriate relationships, including recursive (if any), binary, ternary (if any) and superclass/ subclass (ISA) relationships. **(6 marks)**

(c) Clear participation constraints and cardinalities. **(4 marks)**

(d) Proper indication of primary keys, partial keys, and descriptive attributes. Note: State any assumptions that you make. **(4 marks)**

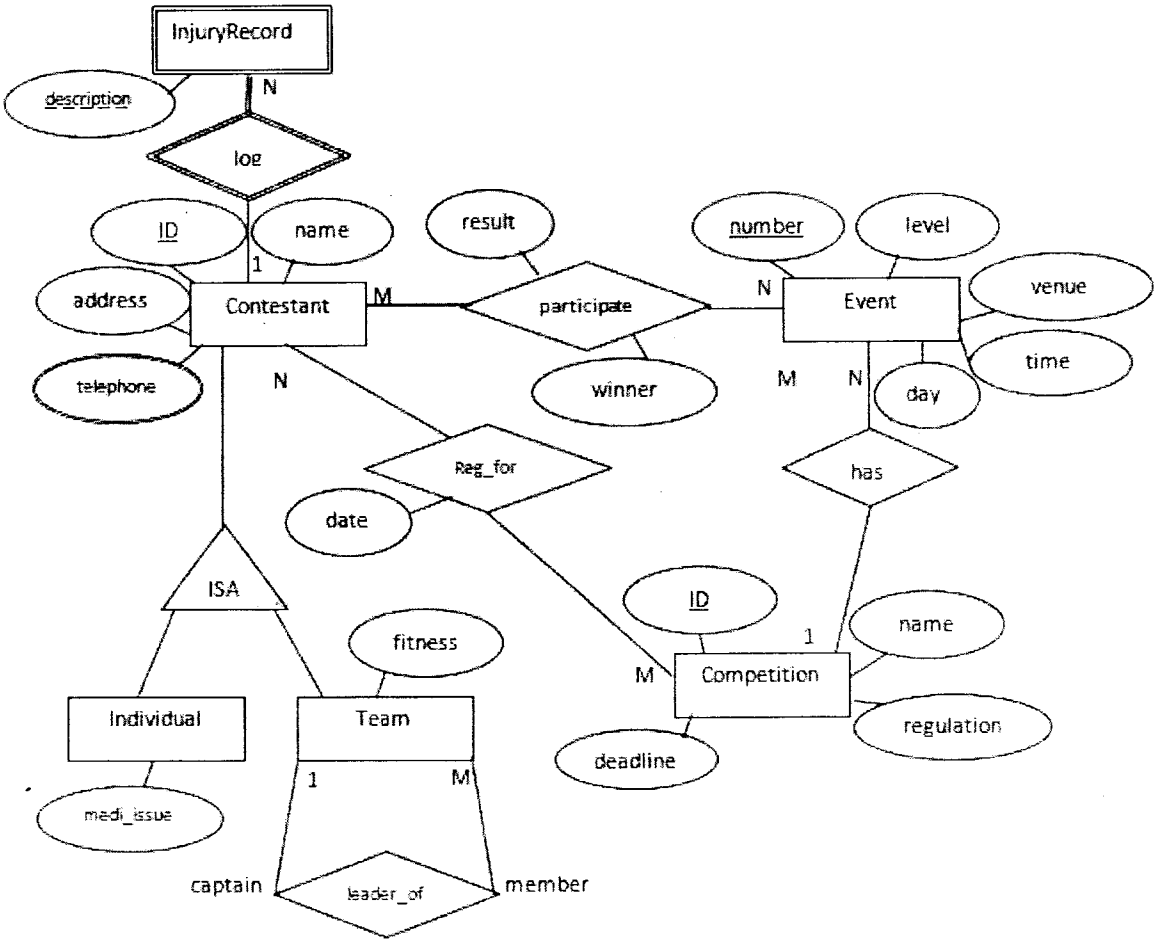
Question 2 (20 marks)

(a) Consider the following EER model.

i. What is the best option for mapping the ISA hierarchy in the diagram? Justify your answer.

(3 Marks)

ii. Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. **(10 Marks)**



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(b) Consider the following relation schema Product storing supermarket product information and functional dependencies FD1, FD2, and FD3.

Product (ProductID, ProductName, CategoryID, CategoryNane, SupplierID, SupplierName)

FD1: ProductID → ProductName, CategoryID, SupplierID

FD2: CategoryID → CategoryNane

FD3: SupplierID → SupplierName

i. Find all the keys using the attribute closure method. **(2 Marks)**

ii. Is the relation in 3NF? Give reasons for your conclusion. If the relation is not in 3NF, convert it to a set of 3NF relations. **(5 Marks)**

Question 3 (20 Marks)

(a) Explain the Isolation property with an example. **(3 Marks)**

(b) Consider the following schedules S1 and S2.

S1: R1(A), W2(A), R3(B), W1(B), R2(C), W3(C)
S2: R2(A), R3(B), W1(C), R1(B), W3(A), R2(C)

Here, Ri(X) means read the item X of transaction i, and Wi(X) means write item X of transaction i.

i. By drawing a precedence graph, identify the schedule that is serializable and that is not serializable. Justify your answer. **(4 Marks)**

ii. Give the possible serial schedule for the identified serializable schedule. **(2 Marks)**

(c) Consider the following interleaved execution involving transactions T1 and T2, where both access and modify data items A and B.

Transaction T1 transfers ₹50 from A to B.

Transaction T2 calculates a 10% interest rate on A and updates it.

T1	**T2**
Read(A)	
	Read(A)
	A = A * 1.10
	Write(A)
Read(A)	
A = A - 50	
Write(A)	
Read(B)	
B = B + 50	
Write(B)	

i. Identify and explain the concurrency control problem that occurs in the above schedule.
(2 Marks)

ii. After applying the Two-Phase Locking (2PL) protocol to prevent the problem identified in (a), an undesirable effect may still occur. Identify this effect and explain how it occurs in this schedule.
(3 Marks)

iii. Explain how 2PL overcomes the concurrency problem identified in part (a).
(3 Marks)

iv. Suppose the same schedule is controlled using the Timestamp Ordering (TO) protocol instead of 2PL. Assume the timestamps are assigned as:
TS(T1) = 5 and TS(T2) = 10 (T1 starts earlier).
For each operation in the schedule, determine whether it executes or causes an abort, and explain why. **(3 Marks)**

Question 4 (20 Marks)

Consider the following relational database schema for a University Library Management system.

The customer relation provides details of the customers. The employee relation holds information about the employees who manage the bookings. The room relation contains details about the rooms available in the hotel. The reservation relation records each room reservation.

Member (mid, mname, gender, city)
Librarian (lid, lname, branch)
Book (bid, title, author, category, price)
Borrow (mid, lid, bid, borrow_date, return_date)

(a) Express each of the following queries as expressions in relational algebra:

i. Retrieve the names of librarians who have processed borrowings of books priced above Rs. 1,000 on 18/07/2024. **(2 Marks)**

ii. List all book IDs that have never been borrowed. **(2 Marks)**

iii. Get the details of books that have been borrowed on both 15/07/2024 and 20/07/2024.

(2 Marks)

iv. Retrieve the names of members who have borrowed all book categories available in the library.

(2 Marks)

v. List each book ID along with the total number of distinct members who have borrowed that book. **(2 Marks)**

(b) Express the above queries (i) to (iv) in tuple relational calculus. **(10 Marks)**

Question 5 (20 Marks)

(a) A file contains r = 25,000 STUDENT records of fixed length. Each record has the following fields:

Name (32 bytes), StudentID (10 bytes), Address (50 bytes), Phone (10 bytes), Birth_date (8 bytes), Sex (1 byte), Dept_code (4 bytes), Year_code (2 bytes), GPA (4 bytes), Status (3 bytes). An additional 1 byte is used as a deletion marker. The file is stored on a disk with a block size = 4096 bytes using spanned organization with a 5-byte pointer at the end of each block. Use the following disk parameters:

Seek time = 25 ms
Transfer rate = 100 MB/s
Disk rotates at 4800 RPM

Answer the following:

i. Calculate the record size R in bytes. **(2 Marks)**

ii. Calculate the average rotational delay of the disk. **(3 Marks)**

iii. Calculate the blocking factor (bfr) of the file. **(3 Marks)**

iv. Calculate the number of blocks required to store the file. **(2 Marks)**

v. If the file is ordered by StudentID, calculate the time to find a record by binary search. **(3 Marks)**

(b)

i. You are given two application scenarios. For each one, identify the most appropriate type of database (Document, Graph, Spatial, Vector). **(2 Marks)**

A. A startup aims to develop a social media platform that enables users to follow one another, post content, and tag posts.

B. A startup wants to design an online e-commerce platform where users can browse products, add items to a shopping cart, and write reviews.

ii. The following table represents a relational model for storing book information in a library system.

Book

BookID	Title	Author	Genre	Price
1	Database Systems	Elmasri	Education	1500
2	AI Basics	Russell	Technology	1800
3	Harry Potter	Rowling	Fiction	1200

A. Explain how the above relational model can be converted into a column-family database model used in NoSQL systems. **(3 Marks)**

B. Explain how data is retrieved in a column-family database compared to a relational database for the following query. **(2 Marks)**

```
SELECT Title
FROM Book
WHERE Genre = 'Technology'
```

---END OF QUESTION PAPER--